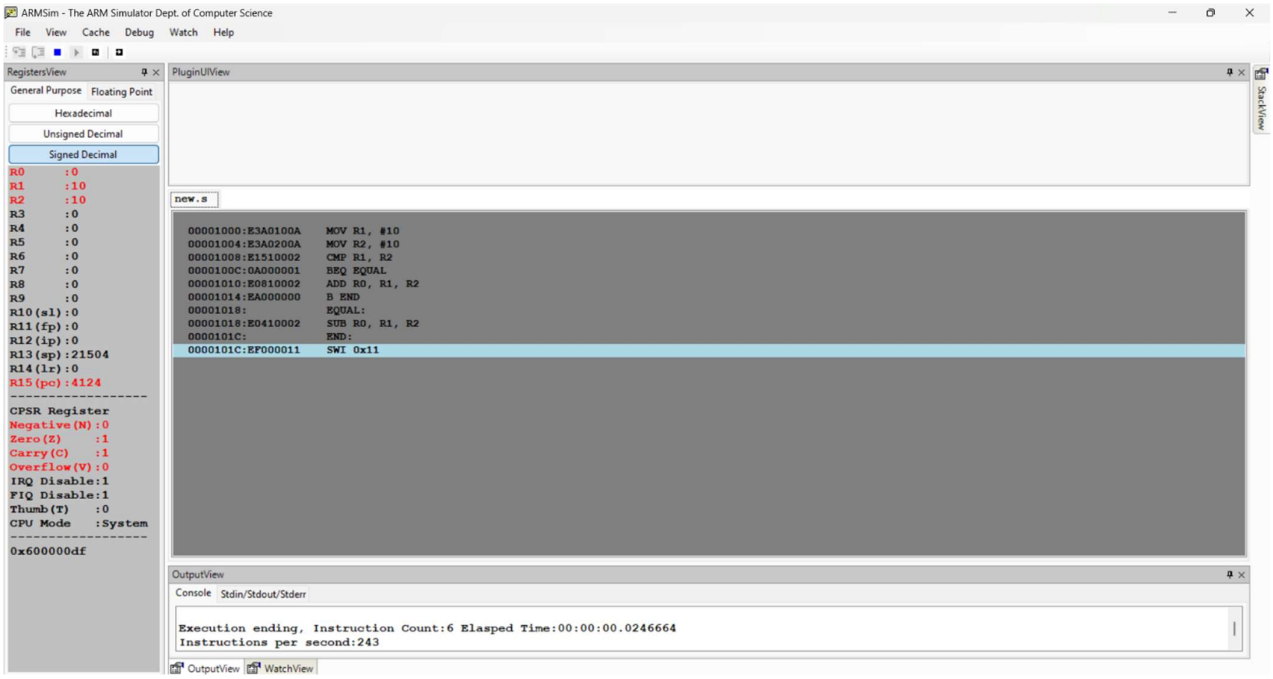
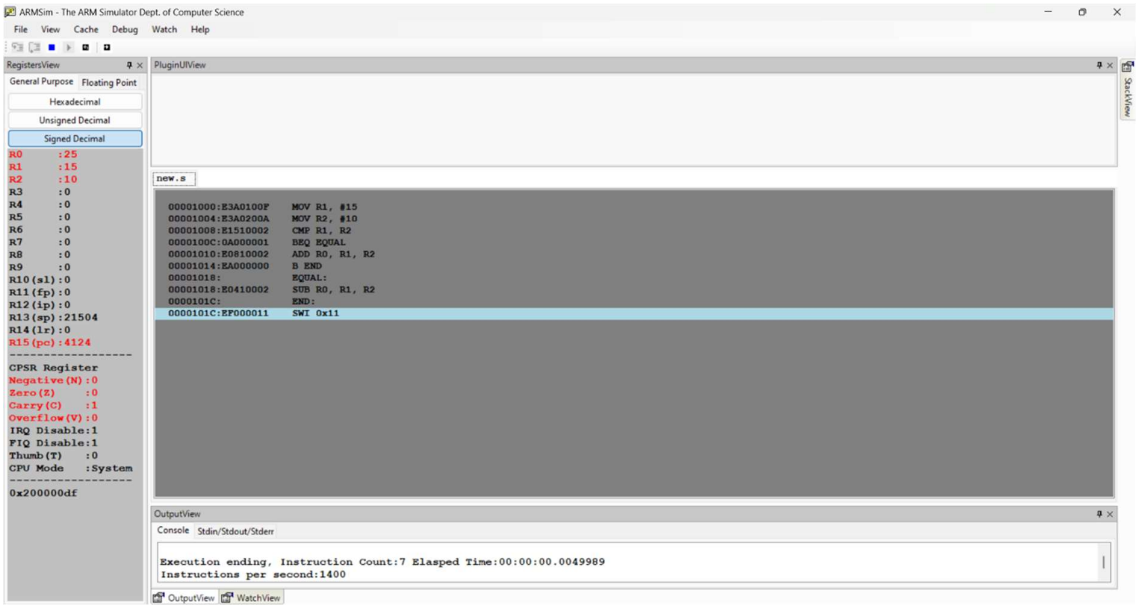
Department of Computer Science & Engineering
Microprocessor & Computer Architecture Lab

Lab 2,3 and 4 Programs
UE24CS251B

Q1. Write an ALP to subtract if the numbers are equal, otherwise add them.

```
MOV R1, #10     @alternatively, I also tried 15 here
MOV R2, #10
CMP R1, R2
BEQ EQUAL
ADD R0, R1, R2
B END
EQUAL:
SUB R0, R1, R2
END:
SWI 0x11
```

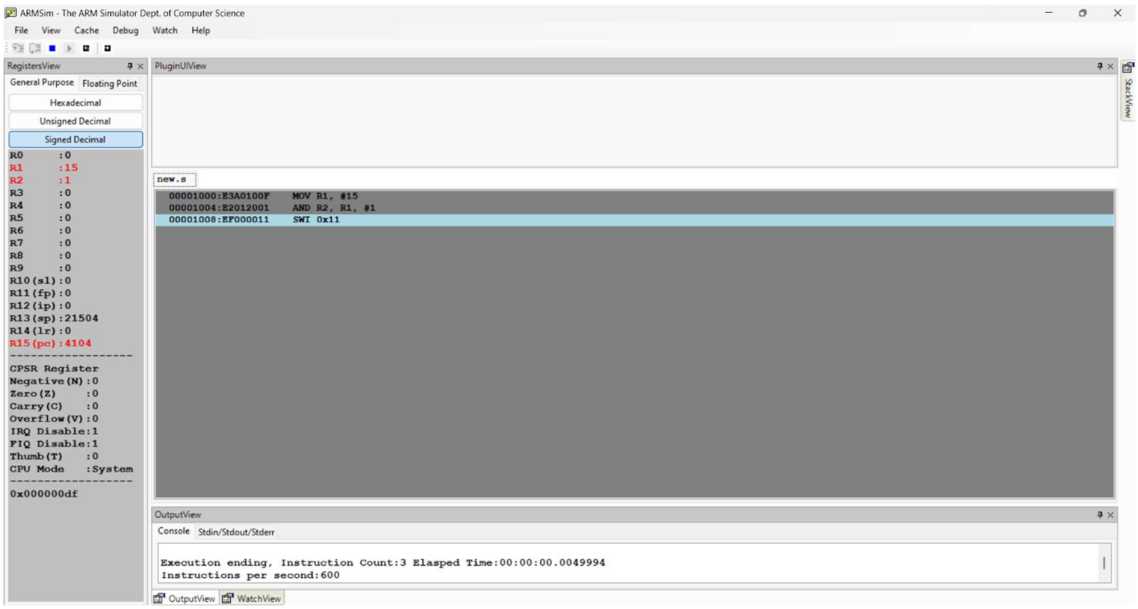


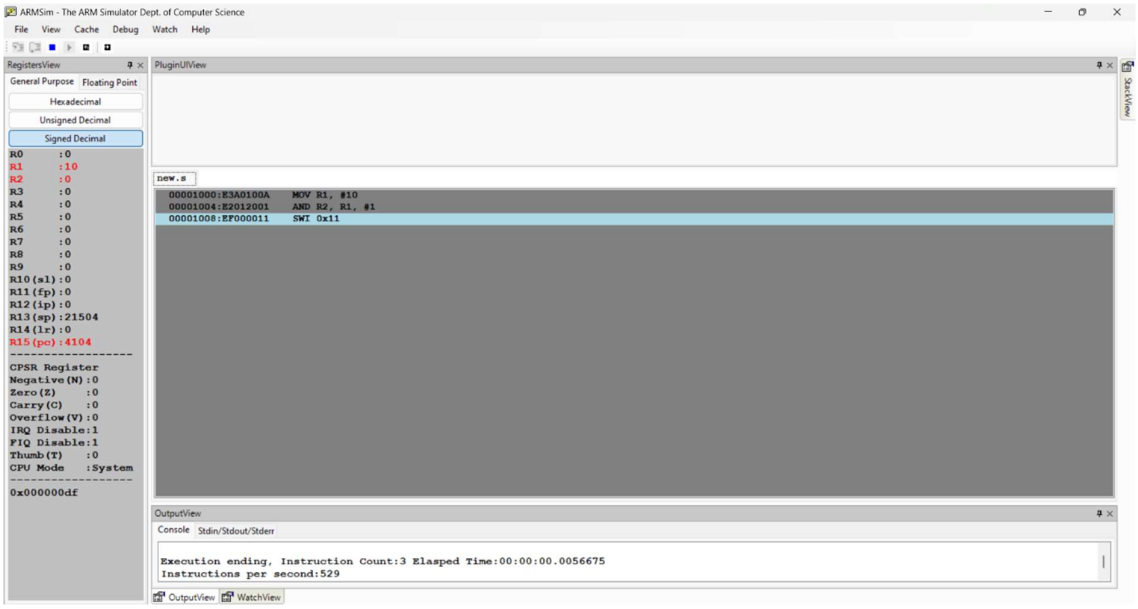


Q2. Check if a given number is even or odd.

Note: at the end of the program execution R2 contains 0 if number is even, otherwise R2 contains 1.

```
MOV R1, #15          @alternatively, I also tried 10 here
AND R2, R1, #1
SWI 0x11
```

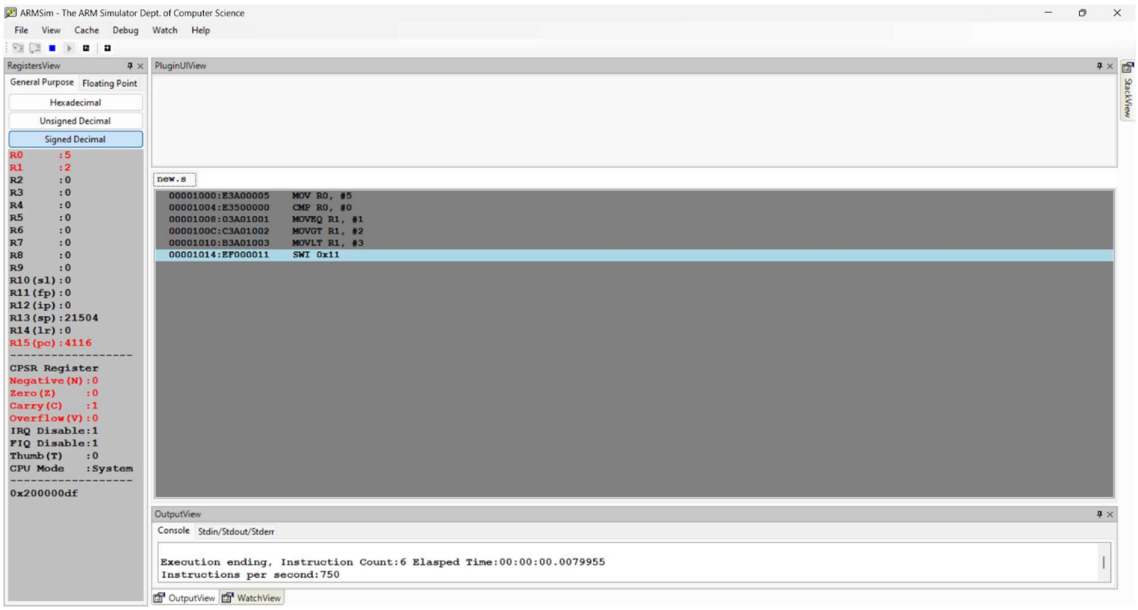


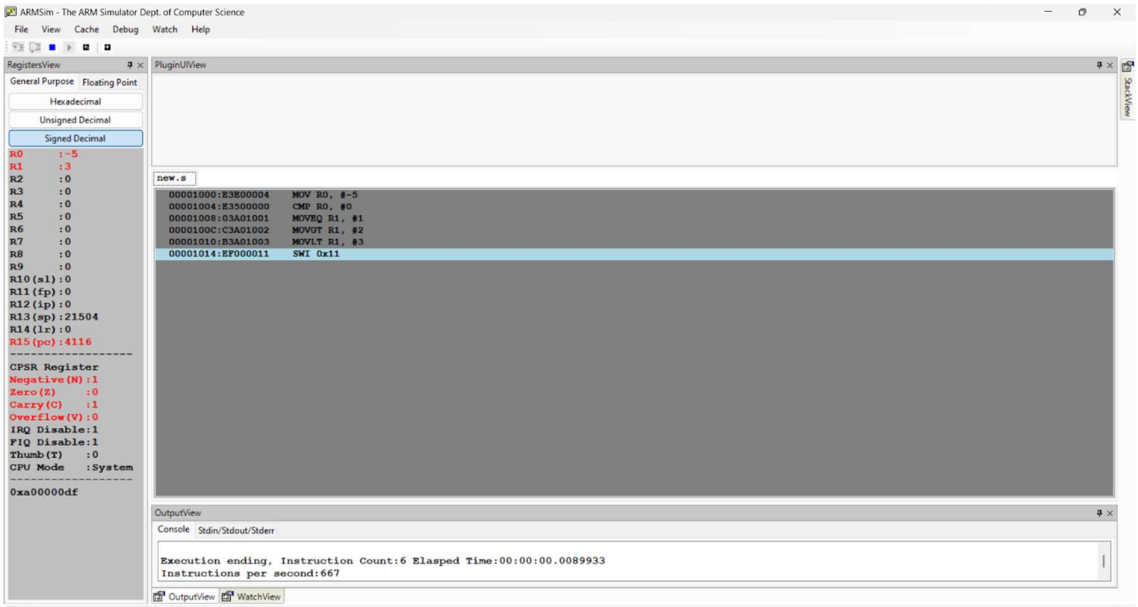


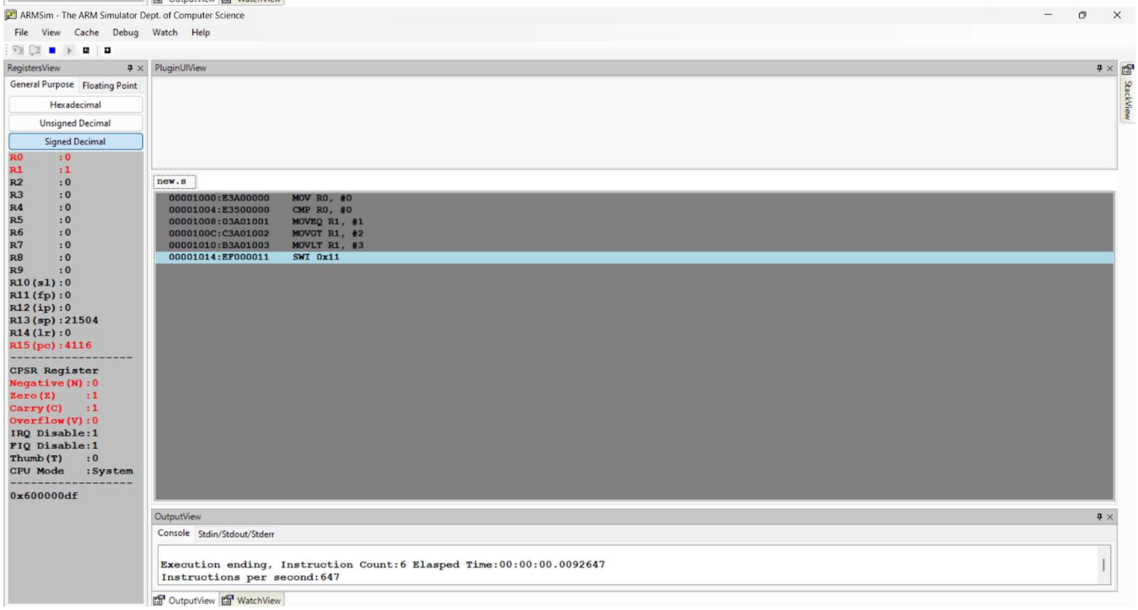
Q3. Based on value of number in r0,

- store 1 in r1 if r0 is 0
- store 2 in r1 if r0 is positive
- store 3 if r0 is negative

```
MOV R0, #5      @alternatively, I also tried -5 and 0 here
CMP R0, #0
MOVEQ R1, #1
MOVGT R1, #2
MOVLT R1, #3
SWI 0x11
```

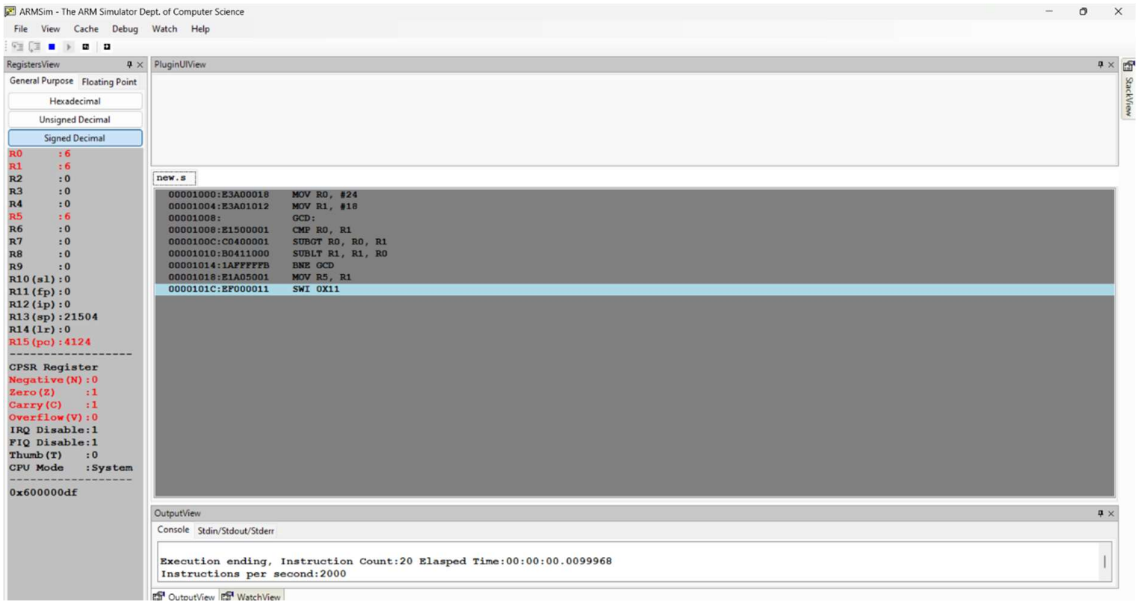






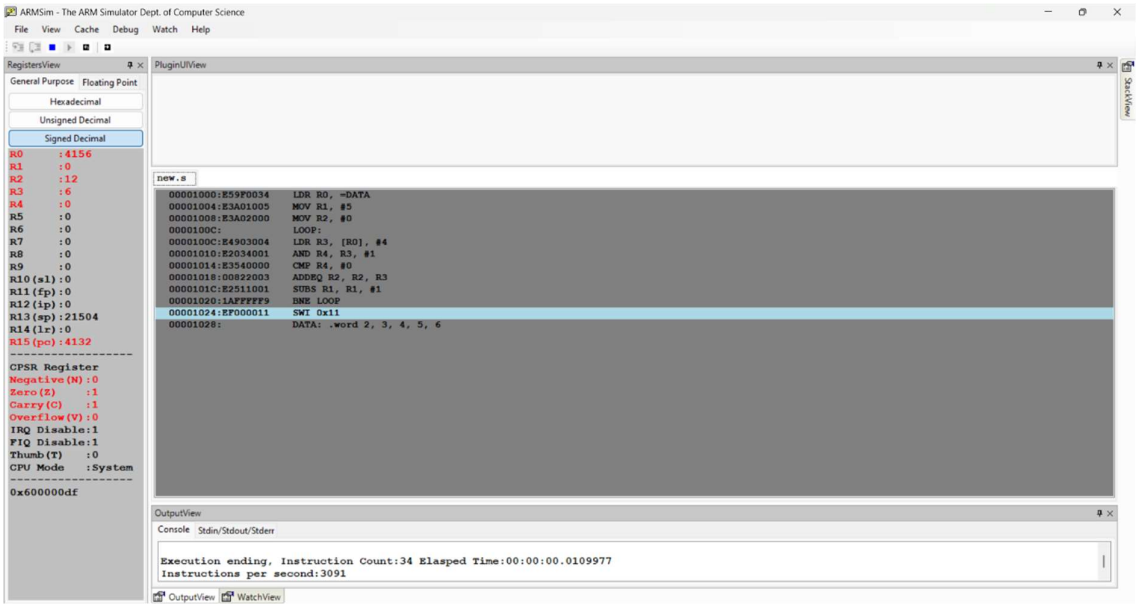
Q4. To find GCD of two numbers.

```
MOV R0, #24
MOV R1, #18
GCD:
CMP R0, R1
SUBGT R0, R0, R1
SUBLT R1, R1, R0
BNE GCD
MOV R5, R1
SWI 0X11
```



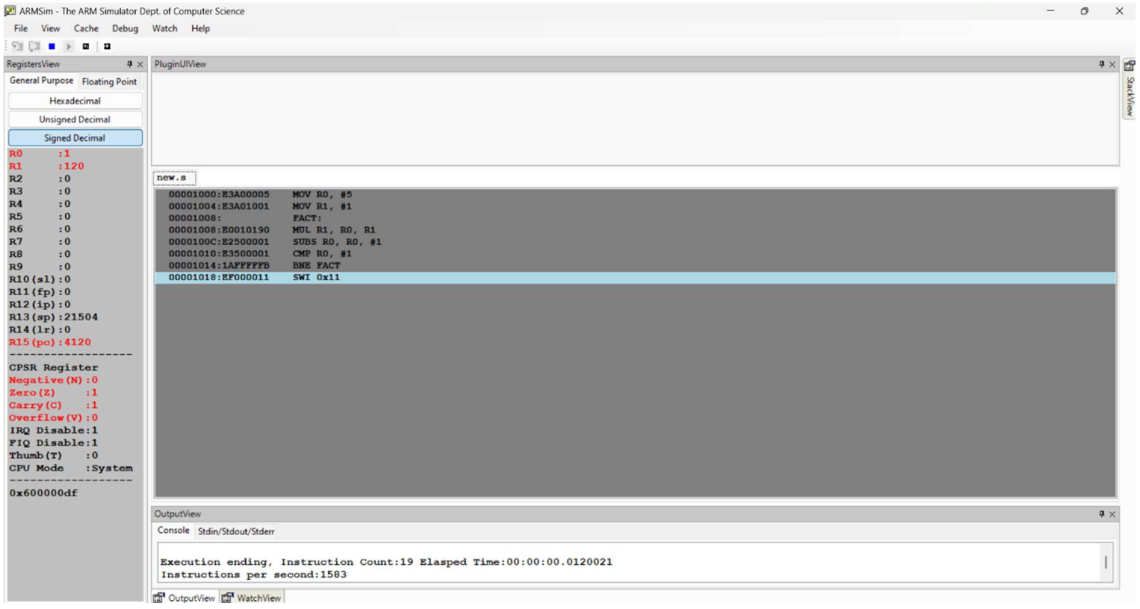
Q5. Write an ALP using ARM7TDMI to add only even numbers stored in memory location for a given set of numbers and store the sum in the memory location.

```
LDR R0, =DATA
MOV R1, #5
MOV R2, #0
LOOP:
LDR R3, [R0], #4
AND R4, R3, #1
CMP R4, #0
ADDEQ R2, R2, R3
SUBS R1, R1, #1
BNE LOOP
SWI 0x11
DATA: .word 2, 3, 4, 5, 6
```



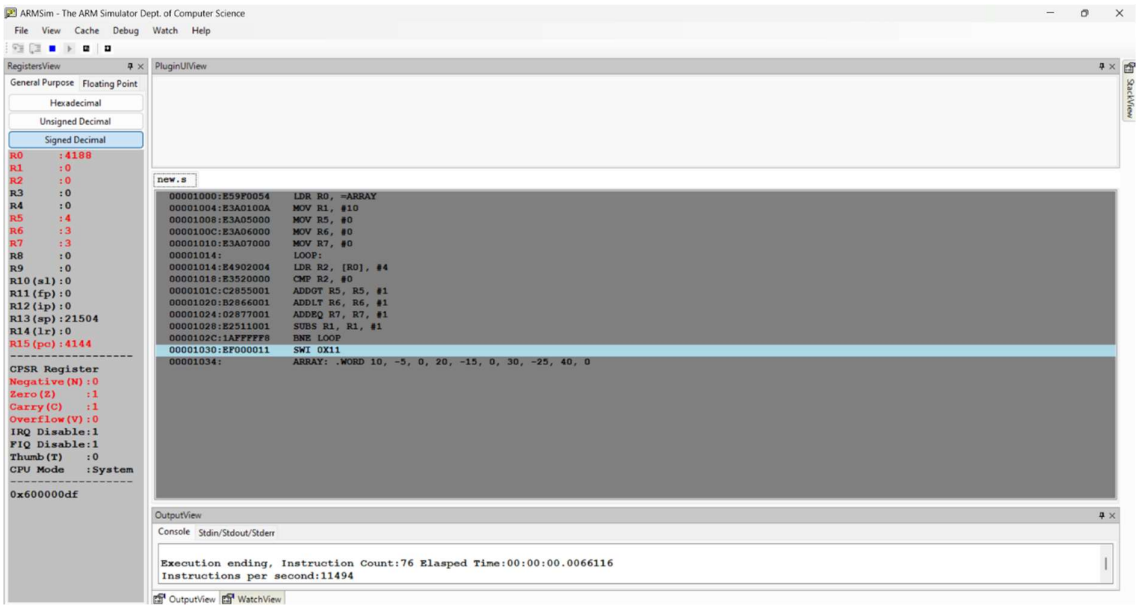
Q6. Write a program to find the factorial of a given number.

```
MOV R0, #5
MOV R1, #1
FACT:
MUL R1, R0, R1
SUBS R0, R0, #1
CMP R0, #1
BNE FACT
SWI 0x11
```



Q7. Write an ALP to find the number of zeroes, positive and negative numbers in a given array

```
LDR R0, =ARRAY
MOV R1, #10
MOV R5, #0
MOV R6, #0
MOV R7, #0
LOOP:
LDR R2, [R0], #4
CMP R2, #0
ADDGT R5, R5, #1
ADDLT R6, R6, #1
ADDEQ R7, R7, #1
SUBS R1, R1, #1
BNE LOOP
SWI 0X11
ARRAY: .WORD 10, -5, 0, 20, -15, 0, 30, -25, 40, 0
```

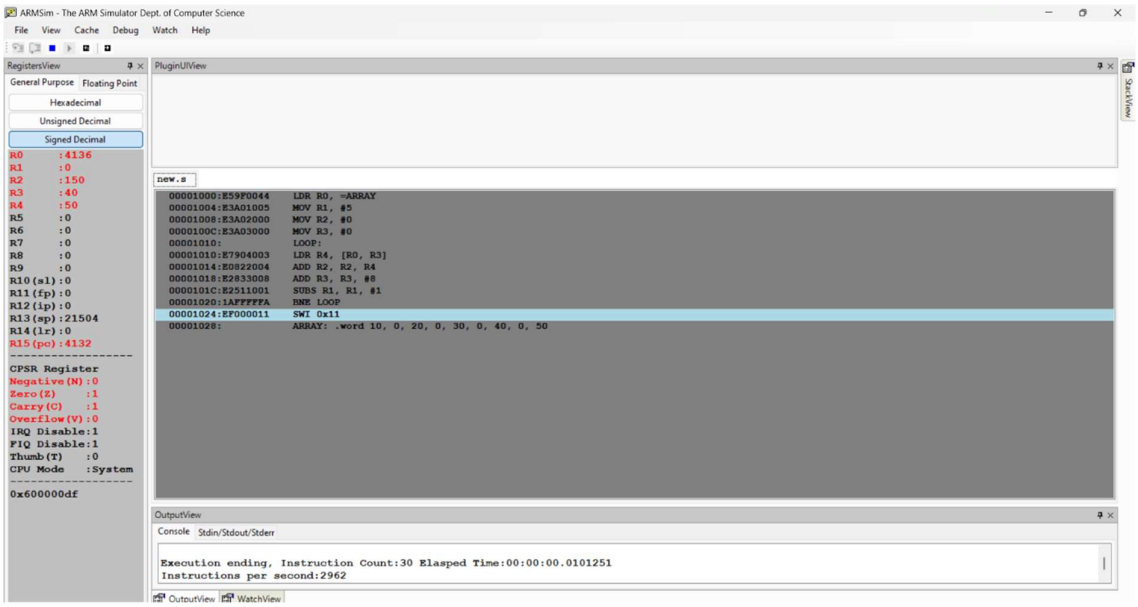


Q8. Write a program in ARM7TDMI-ISA to find the sum of N data items at alternate locations in the memory. Store the result in the memory location.

- Use Pre-indexing addressing mode
- Use Post-indexing addressing mode
- Use Auto-indexing addressing mode

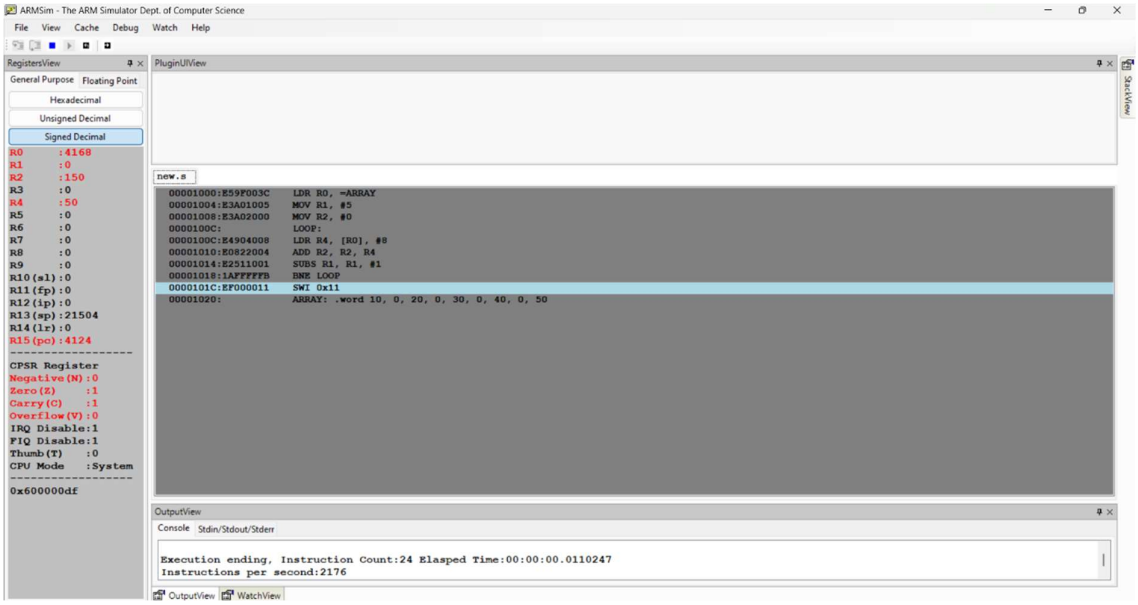
Pre indexed:

LDR R0, =ARRAY
MOV R1, #5
MOV R2, #0
MOV R3, #0
LOOP:
LDR R4, [R0, R3]
ADD R2, R2, R4
ADD R3, R3, #8
SUBS R1, R1, #1
BNE LOOP
SWI 0x11
ARRAY: .word 10, 0, 20, 0, 30, 0, 40, 0, 50



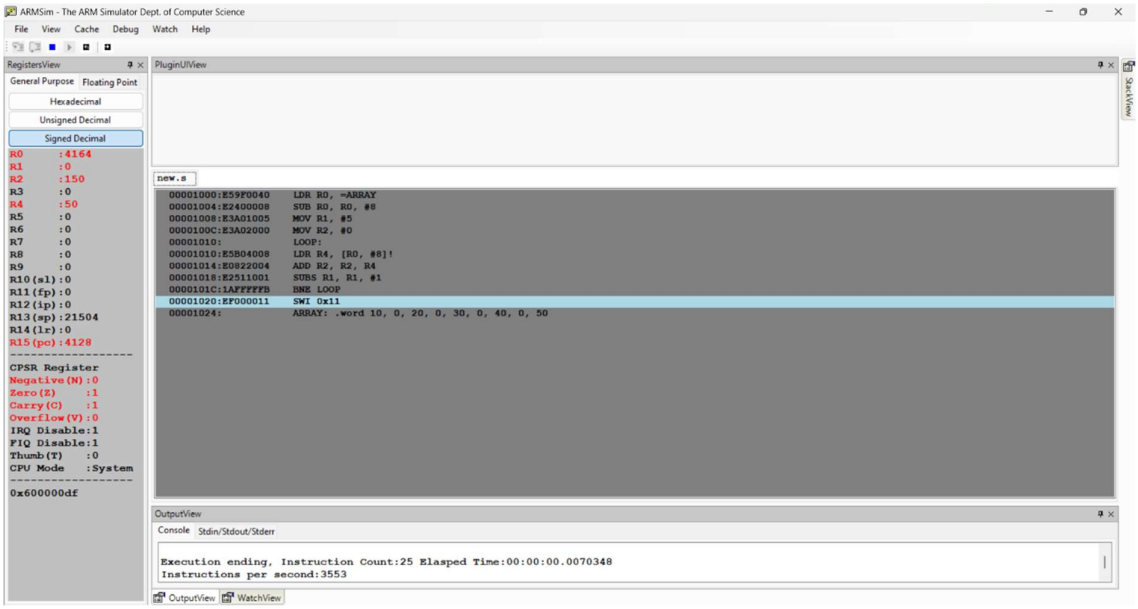
Post indexed:

LDR R0, =ARRAY
MOV R1, #5
MOV R2, #0
LOOP:
LDR R4, [R0], #8
ADD R2, R2, R4
SUBS R1, R1, #1
BNE LOOP
SWI 0x11
ARRAY: .word 10, 0, 20, 0, 30, 0, 40, 0, 50



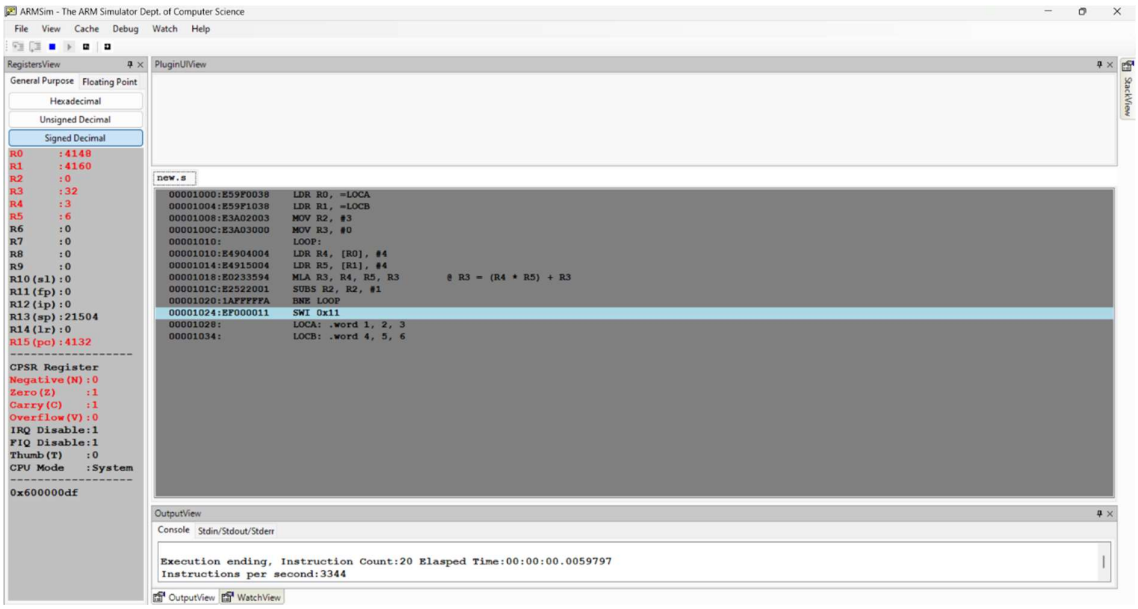
Auto indexed:

LDR R0, =ARRAY
SUB R0, R0, #8
MOV R1, #5
MOV R2, #0
LOOP:
LDR R4, [R0, #8]!
ADD R2, R2, R4
SUBS R1, R1, #1
BNE LOOP
SWI 0x11
ARRAY: .word 10, 0, 20, 0, 30, 0, 40, 0, 50



Q9. Write an ALP to perform Convolution using MUL and MLA instruction (Addition of multiplication of respective numbers of loc A and loc B).

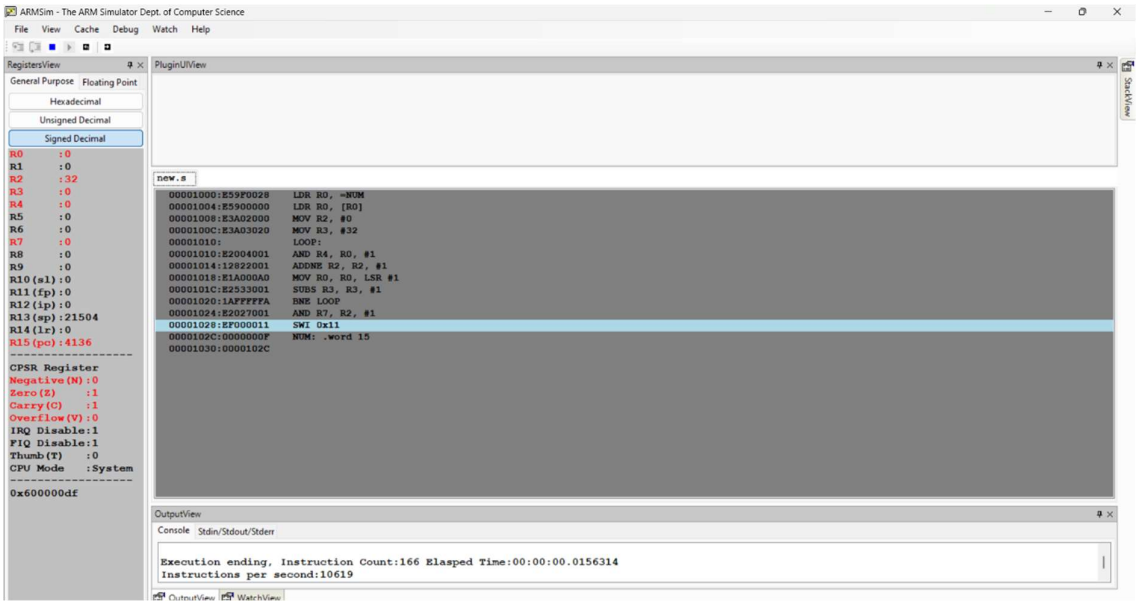
```
LDR R0, =LOCA
LDR R1, =LOCB
MOV R2, #3
MOV R3, #0
LOOP:
LDR R4, [R0], #4
LDR R5, [R1], #4
MLA R3, R4, R5, R3      @ R3 = (R4 * R5) + R3
SUBS R2, R2, #1
BNE LOOP
SWI 0x11
LOCA: .word 1, 2, 3
LOCB: .word 4, 5, 6
```



Q10. Write an ARM7TDMI ALP to check whether a given 32-bit number has odd or even parity (number of 1s).
It stores 00 in R7 if parity is EVEN and 01 in R7 if parity is ODD

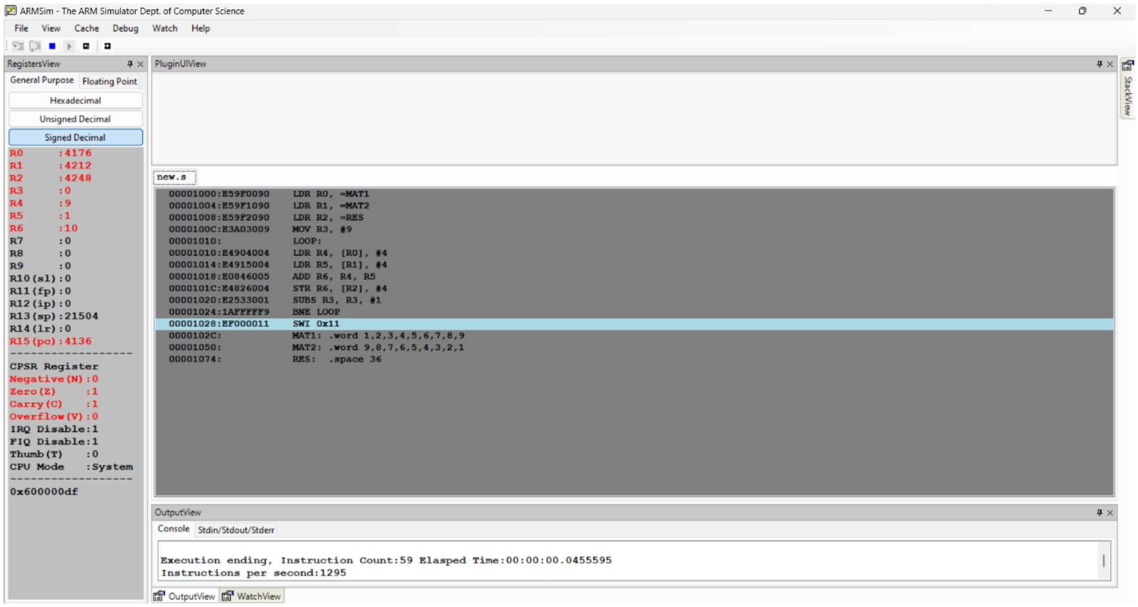
```
LDR R0, =NUM
LDR R0, [R0]
MOV R2, #0
MOV R3, #32
LOOP:
AND R4, R0, #1
```

ADDNE R2, R2, #1
MOV R0, R0, LSR #1
SUBS R3, R3, #1
BNE LOOP
AND R7, R2, #1
SWI 0x11
NUM: .word 15



Q11. Write an ARM ALP to perform element-wise addition of two 3×3 matrices using indexed addressing

LDR R0, =MAT1
LDR R1, =MAT2
LDR R2, =RES
MOV R3, #9
LOOP:
LDR R4, [R0], #4
LDR R5, [R1], #4
ADD R6, R4, R5
STR R6, [R2], #4
SUBS R3, R3, #1
BNE LOOP
SWI 0x11
MAT1: .word 1,2,3,4,5,6,7,8,9
MAT2: .word 9,8,7,6,5,4,3,2,1
RES: .space 36


ARMSim - The ARM Simulator Dept. of Computer Science
File View Cache Debug Watch Help
RegistersView
General Purpose Floating Point
Hexadecimal
Unsigned Decimal
Signed Decimal
R0 :4176
R1 :4212
R2 :4248
R3 :0
R4 :9
R5 :1
R6 :10
R7 :0
R8 :0
R9 :0
R10(sl):0
R11(fp):0
R12(ip):0
R13(sp):21504
R14(lr):0
R15(pc):4136
CPSR Register
Negative(N):0
Zero(Z) :1
Carry(C) :1
Overflow(V):0
IRQ Disable:1
FIQ Disable:1
Thumb(T) :0
CPU Mode :System
0x600000df
PluginUIView
StackView
new.s
00001000:E59F0090 LDR R0, =MAT1
00001004:E59F1090 LDR R1, =MAT2
00001008:E59F2090 LDR R2, =RES
0000100C:E3A03009 MOV R3, #9
00001010: LOOP:
00001010:E4904004 LDR R4, [R0], #4
00001014:E4915004 LDR R5, [R1], #4
00001018:E0846005 ADD R6, R4, R5
0000101C:E4826004 STR R6, [R2], #4
00001020:E2533001 SUBS R3, R3, #1
00001024:1AFFFFF9 BNE LOOP
00001028:EF000011 SWI 0x11
0000102C: MAT1: .word 1,2,3,4,5,6,7,8,9
00001050: MAT2: .word 9,8,7,6,5,4,3,2,1
00001074: RES: .space 36
OutputView
Console Stdin/Stdout/Stderr
Execution ending, Instruction Count:59 Elasped Time:00:00:00.0455595
Instructions per second:1295
OutputView WatchView